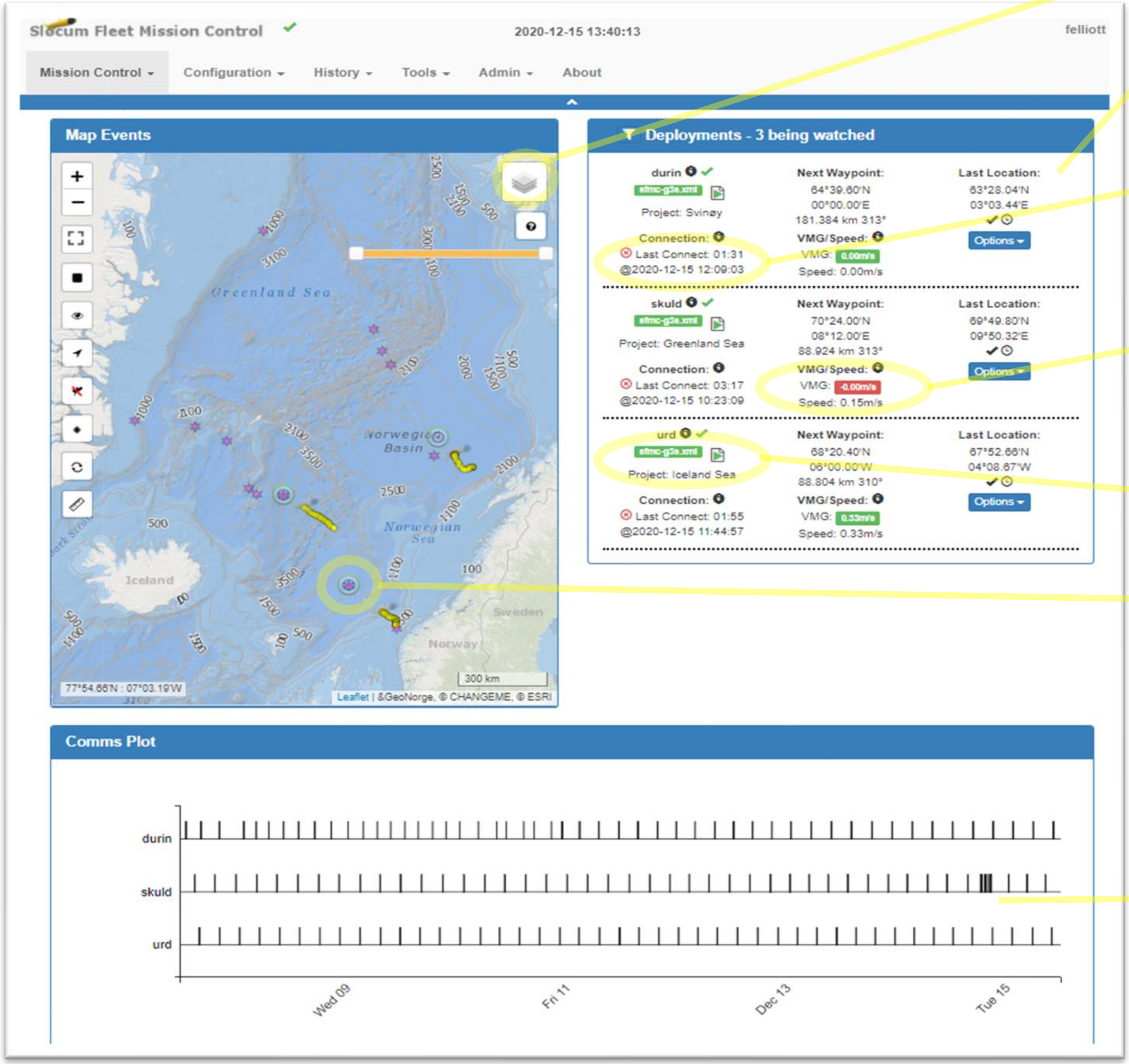
Dashboard Summary



Toggle map layers here

Click on deployment to highlight tracks in map

CHECKLIST:

❑ **Glider has called in within the last 6hrs**

It's unusual for gliders to miss call-in's but it does happen in rough conditions. If a glider has not called in after 14hrs inform glider team.

❑ **Glider is making positive progress**

Gliders can be easily overwhelmed in strong currents, check glider is not being pushed too far off course or in to hazards. Zoom in on glider tracks to see progress. If strong currents persist after 14hrs adjust waypoints.

❑ **XML script is enabled**

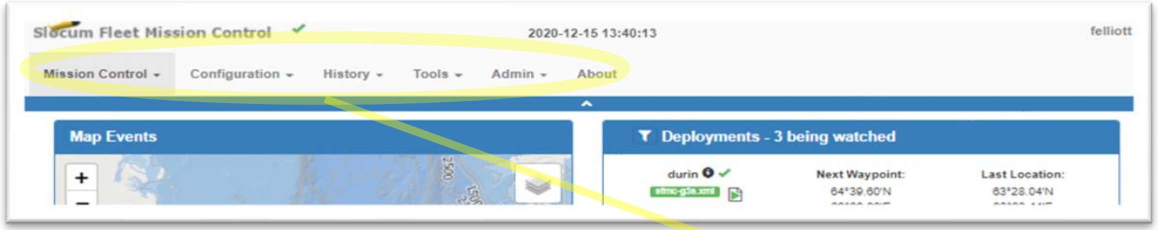
Pilot often forget to re-enable scripts after manually talking to gliders.

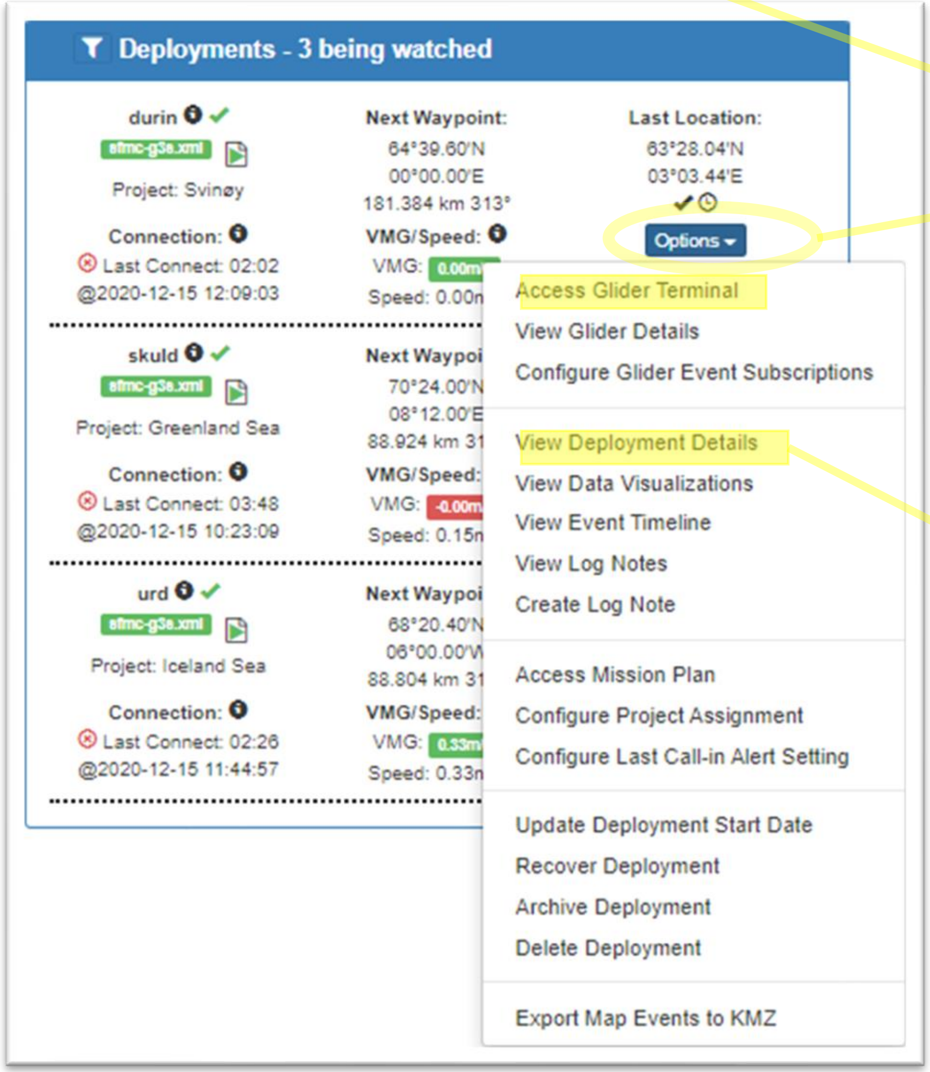
❑ **Current waypoint**

Is this correct? It is very common for to waypoint list to get reloaded and the glider to start heading for a previously achieved waypoint (see waypoint handling on how to avoid this)

The *Comms Plot* is a visual representation of glider call in times, we try keep consistent surfacing interval of 5.5hrs in a full dive depth of 1000m.

Dashboard Site Navigation





NOTES:

- Click the *Options* button for deployment specific links or the navigation bar for generic page links, spend time following all these links to become familiar with them.
- *Dashboard, Access Glider Terminal* and *View Deployment Details* are the two pages you will use most

CHECKLIST:

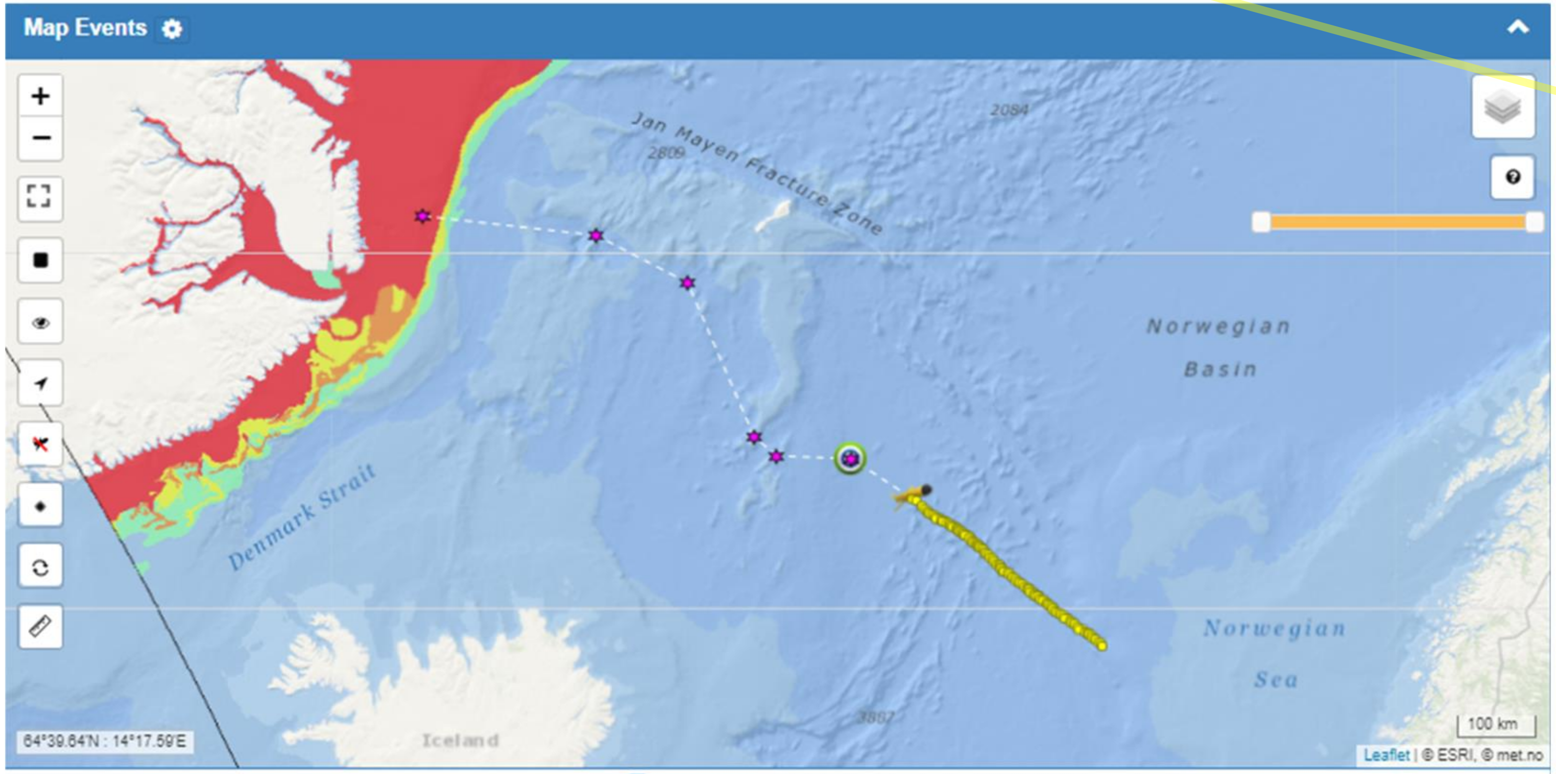
- [ ] Click on *View Deployment Details* for a deployment

Or if there is a glider currently connected you may want to go to that first.

Deployment Details – Summary and Map

Summary

Glider	Time Disconnected (HH:mm)	Deployed	Last Location	Reported Next Waypoint	Range Bearing	Total Distance	Mission	Segment	Script	Project
skuld	01:11	2020-12-04 03:11	69°50.394'N 9°53.136'E	70°24.00'N 8°12.00'E	89.403km 312°	167.43km	uib02a.mi	0037.0012 skuld-2020-348-1-12	sfmc-g3a.xml	Greenland Sea



Hide unselected deployments

CHECKLIST:

- ❑ Expected mission is running, usually uib02a.mi

If a glider aborts or finishes a mission and is not provided further instructions it will eventually sequence laspgasp.mi and continue to run this mission forever.

- ❑ A project is assigned

Not critical but useful for organizing missions and waypoints.

- ❑ Distance to next waypoint

It is easy to send the glider to the wrong hemisphere

NOTES:

- Click on buttons on the map to become familiar with them
- Bathy layers are *Dybdekontur oversiktsdata* & *Dybdekontur label.*
- See *Map Settings for adding layers to SFMC*

Deployment Details – Surfacings

Surfacings

Location	Connection Start	Connection End	Connection Duration	Time Since Prior	Surface Reason	Speed & Distance	Next Waypoint	Range and Bearing	Mission	Mission Segment	Device Status (t/m/s)
69°51.068'N 9°57.951'E	2020-12-16 06:29:13	2020-12-16 06:32:59	00:03:46	00:00	no comms for a while	0.00m/s 0.00km	70°24.00'N 8°12.00'E	90.74km 310°	uib02a.mi	0037.0020	2/0/0 267/16/3 880/77/11
69°51.068'N 9°57.951'E	2020-12-16 06:25:30	2020-12-16 06:28:23	00:02:53	03:58	no comms for a while	0.01m/s 0.11km	70°24.00'N 8°12.00'E	90.67km 310°	uib02a.mi	0037.0019	2/0/0 267/16/3 880/77/11
69°51.129'N 9°57.953'E	2020-12-16 02:22:57	2020-12-16 02:26:39	00:03:42	00:01	no comms for a while	0.00m/s 0.00km	70°24.00'N 8°12.00'E	90.67km 310°	uib02a.mi	0037.0018	2/0/0 264/13/1 869/66/5
69°51.129'N 9°57.953'E	2020-12-16 02:18:32	2020-12-16 02:21:56	00:03:24	03:51	no comms for a while	0.10m/s 1.35km	70°24.00'N 8°12.00'E	89.98km 311°	uib02a.mi	0037.0017	2/0/0 264/13/1 869/66/5
69°50.929'N 9°55.917'E	2020-12-15 22:21:17	2020-12-15 22:27:00	00:05:43	03:47	no comms for a while	0.09m/s 1.22km	70°24.00'N 8°12.00'E	89.98km 311°	uib02a.mi	0037.0015	2/0/0 263/12/1 864/61/5

← First « 1 2 3 » Last →

NOTE: link to log file from surfacing – useful!

CHECKLIST:

❑ Is average connection duration < 7min

Ideally < 5min. If it is consistently too long reduce the tbd and sbd lists.

❑ Is the average time since prior ~5.5hrs

Consistent surfacing intervals and not too many call drops (indicated by quick *Time Since Prior*)

❑ What is the surfacing reason?

Should be Pitch not commanded

pitch not commanded *Pitch not commanded: completed specified number of yo's, yo14.ma*

no comms for a while *No comms for a while: surfacing interval expired, surfac21.ma*

Hit a waypoint *Hit a waypoint: waypoint reached*

MS_ABORT_DEVICE_ERROR Abort error

❑ Note speed & distance trends

Speed and distance during the segment gives an indication of how the glider is performing in the local environment and local currents and tides.

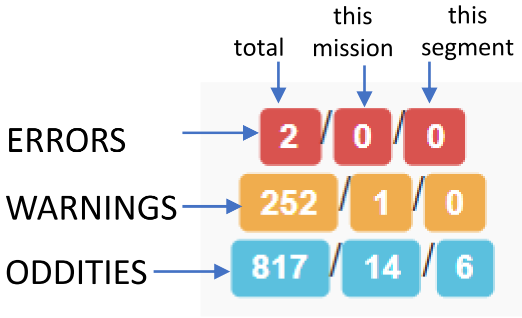
❑ Device Status – important!

Note errors, warnings and oddities occurring. See *Device Status* on following page

Deployment Details – Device Status

Location	Connection Start	Connection End	Connection Duration	Time Since Prior	Surface Reason	Speed & Distance	Next Waypoint	Range and Bearing	Mission	Mission Segment	Device Status (t/m/s)
69°47.628'N 9°45.519'E	2020-12-14 22:10:50	2020-12-14 22:13:59	00:03:09	01:02	pitch not commanded	0.36m/s 1.34km	70°24.00'N 8°12.00'E	89.87km 317°	uib02a.mi	0037.0001	2/0/0 252/1/0 817/14/6

```
name in ALLCAPS means CRITICAL device (* => SUPERCRITICAL)
[I Installed] [- Not_Installed]
[u In_use] [- Not_In_use] [X Out_of_Service]
             name       limits      stats (#total/#mission/#segment)
                      os w/s w/m os     errors        warnings       oddities
 0         simdrvr  -
 1     test_driver  -
 2          ARGOS* I u  -1  20   5  0
 3        WATCHDOG  I u  -1  -1  -1  0
 4         DEADMAN  I u  -1  20   5  0
 5         CONSOLE* I u  -1  20   5  0
 6             GPS  I u  -1  20   5  0 [   0   0  0] [   1   1  0] [   0   0  0]
 7    attitude_rev  I u   3  20   5  0
 8  ocean_pressure  I u   3  20   5  0
 9          vacuum  I u   3  20   5  0
10         battery  I u   3  20   5  0
11        air_pump  I u   3  20   5  0
12     pitch_motor  I u   3  20   5  0 [   0   0  0] [   0   0  0] [   1   0  0]
13   science_super  I u   3  20   5  0
14         digifin  I u   3  20   5  1 [   1   0  0] [  16   0  0] [ 210   1  1]
15       altimeter  I u   3  20   5  0
16        IRIDIUM* I u  -1  20   5  0 [   0   0  0] [   0   0  0] [ 114  12  4]
17      leakdetect  I u   3  20   5  0
18        recovery  I u   3  20   5  0
19         coulomb  I u   3  20   5  0 [   0   0  0] [   0   0  0] [  12   1  1]
20        veh_temp  I u   3  20   5  0
21   BUOYANCY_PUMP  -
22         DE_PUMP  -
23         HD_PUMP  I u   3  20   5  1 [   1   0  0] [ 235   0  0] [ 480   0  0]
24        thruster  I u   3  20   5  0
devices:(t/m/s) errs:   2/   0/   0 warn: 252/   1/   0 odd: 817/  14/   6
```



CHECKLIST:

- ❑ Are there > 5 oddities and/or > 1 warnings?

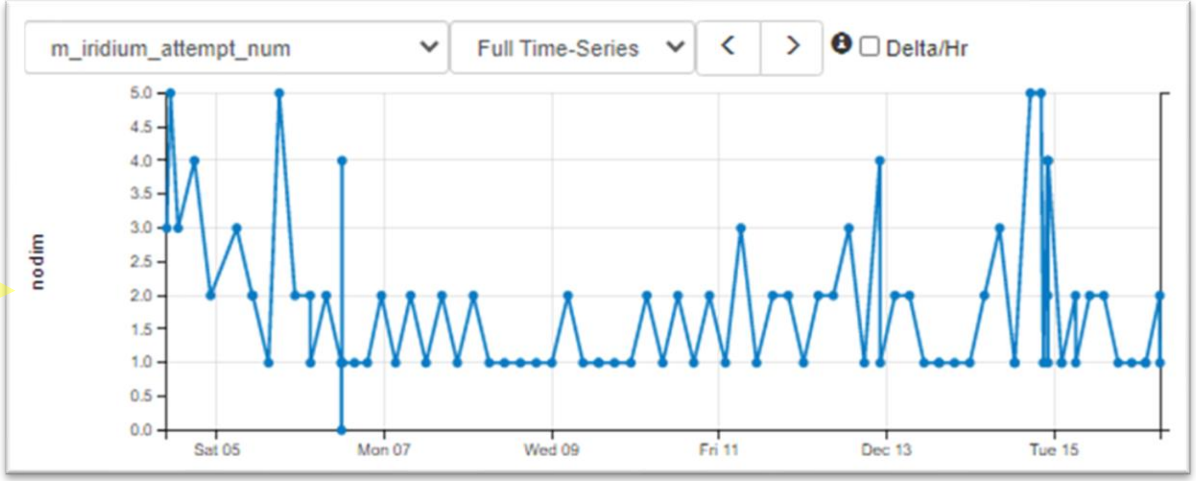
- ❑ Download and open the log file

Scroll down to the bottom of file where the device status has been recorded (sometimes the call drops before this is transmitted)

- ❑ What devices are causing e/w/o's?
  - 1-6 iridium oddities is normal, more is an indication of rough seas, more in fine conditions may indicate poor satellite coverage or a faulty iridium modem.
  - 1-3 digifin oddities are normal, this typical happens at the inflection and on the surface due to wave slap.
  - Sporadic GPS, pitch, science and pump oddities will occur, numerous and consistent oddities or warnings require further investigation.
  - Consider downloading the relevant mlg file.

    s <segment_number>.mlg

- ❑ Plot relevant sensors in *Surface Sensor Values*



Deployment Details – Surface Sensor Values

NOTE: Values available here are from the sensor surface sensor dialog (defined in config.srf file and output from *where* command)

CHECKLIST:

❑ Plot out each sensor

All of these don't need to be checked every day but keep an eye on key sensors or devices that are producing oddities.

❑ Key sensors:

m_avg_upward_inflection_time	Below < 80s, monitor for pump performance
m_battery	Monitor voltage, check its not dropping to fast
m_coulomb_amphr_total	Power consumption, aim for < 3Ah
m_bms_aft_current m_bms_ebay_current m_bms_pitch_current	BMS
m_vacuum	Dropping vacuum can indicate slow leak
m_avg_climb_rate m_avg_dive_rate	Indicates glider ballasting, biofouling
c_climb_bpump c_dive_bpump	Indicates glider ballasting, autoballast performance
c_autoballast_state c_autoballast_volume	Autoballast settings
m_iridium_attempt_num m_iridium_signal_strength	Iridium health
m_leakdetect_voltage m_leakdetect_voltage_forward m_leakdetect_voltage_science m_digifin_leakdetect_reading	Leak detect indication, these will trigger aborts before you notice.

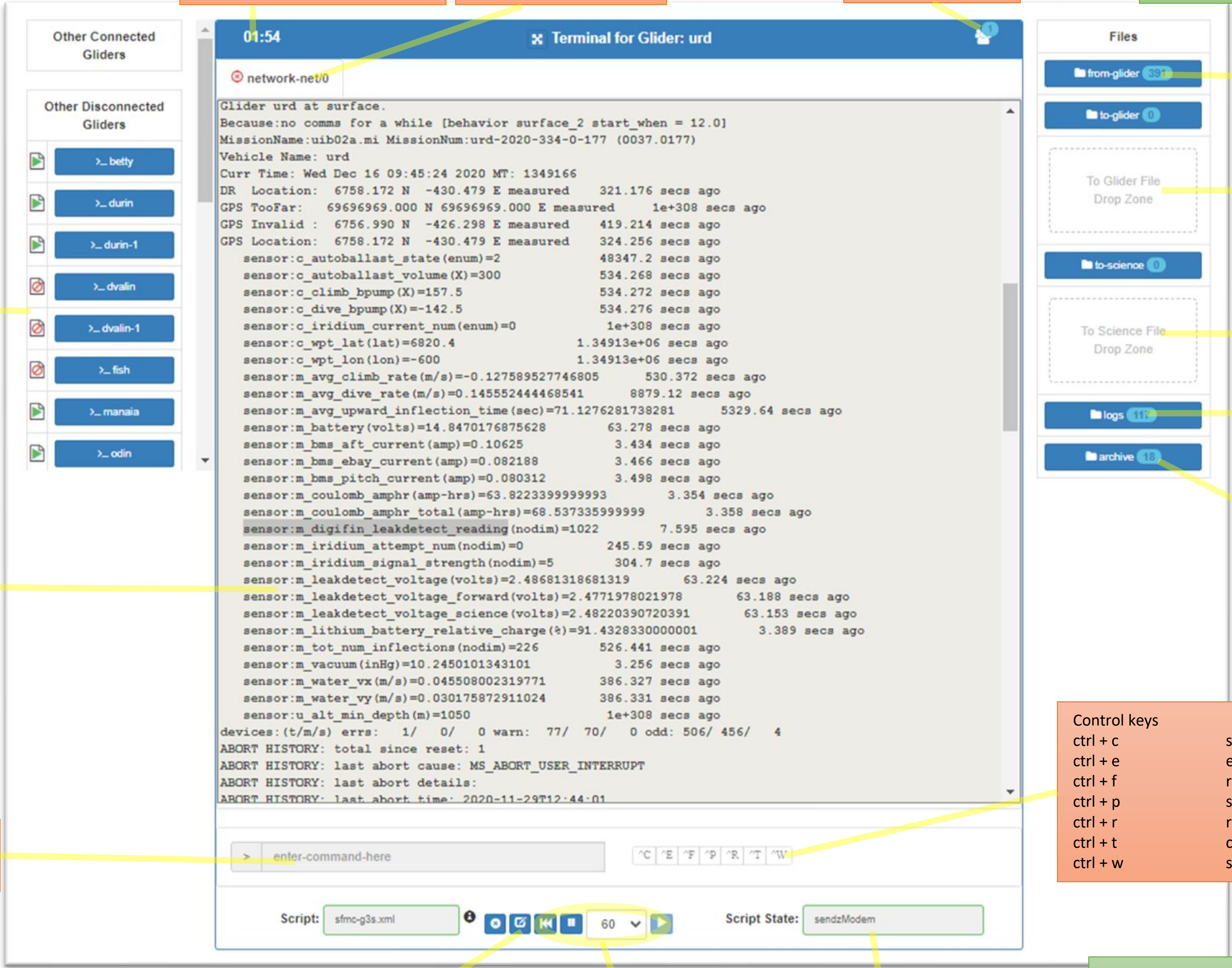
Glider Terminal - overview

How long glider has been connected/disconnected

Connection type (rudics/modem) and port number

Users logged in

NOTE: poor internet connections will not automatically refresh the page, you may need to do it manually



Files sent from the glider

To-glider folder - Files being sent to glider main processor dropped here

To-science folder - Files being sent to glider main processor dropped here

Log files from glide surfacing

Archive - Copies of previously sent files can be downloaded from the *archive* folder

Access other glider terminals

Terminal window

Enter glider commands here, remember to pause script first

Control keys	
ctrl + c	stop mission
ctrl + e	extend surface time by 5mi
ctrl + f	re-read mafiles
ctrl + p	start mission immediately
ctrl + r	resume mission
ctrl + t	consci
ctrl + w	show device status

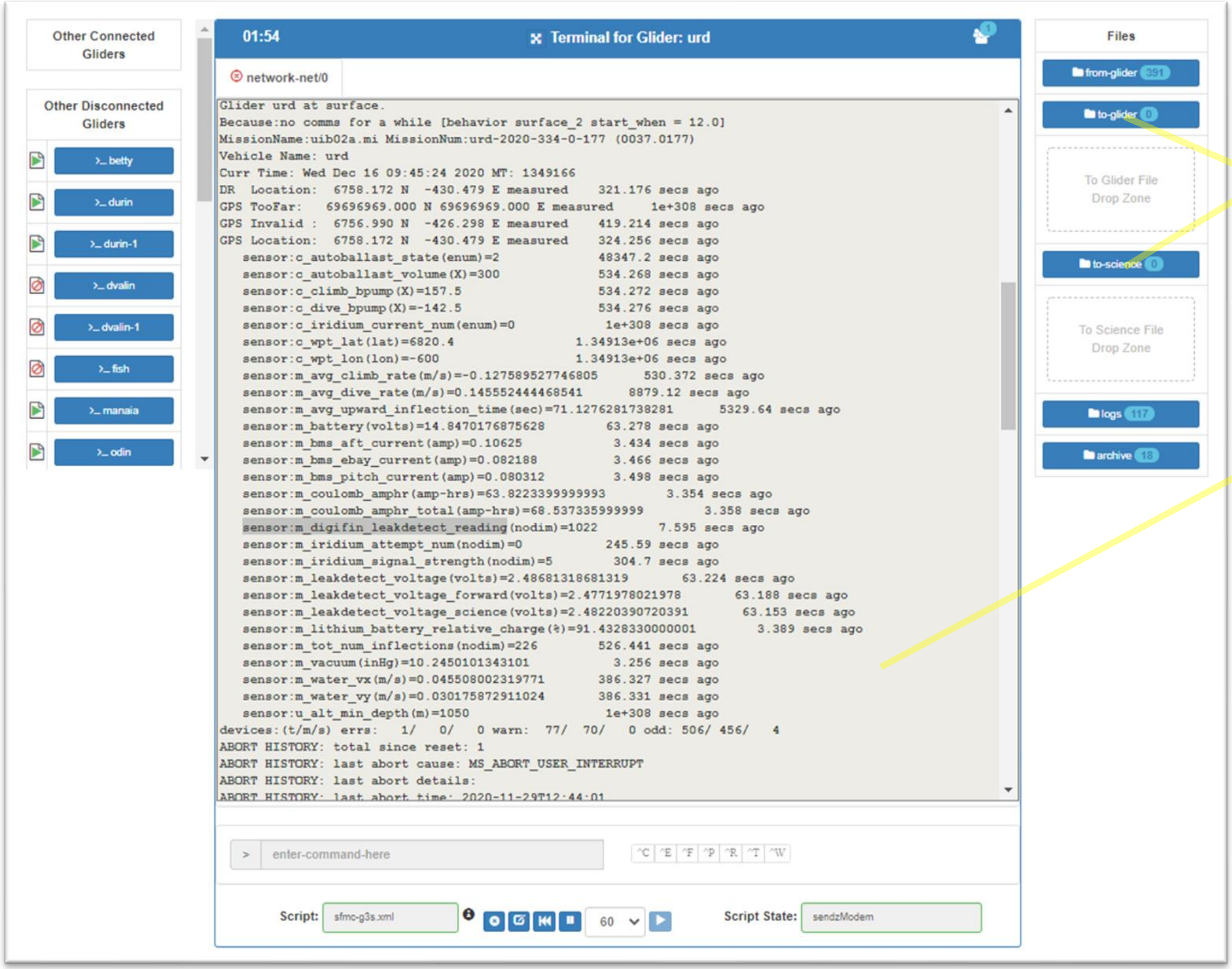
Select XML script

Script controls

Script state

NOTE: State of XML script, when call drop the script can sit in the wrong state and may need to be *rewinded*

Glider Terminal



CHECKLIST:

- ❑ How long has the glider been connected or disconnected?

- ❑ Are files present in to-glider, to-science?

Sometimes the xml script will not transfer files, reset the script or transfer file manually

- ❑ Drop files to be sent to glider in *to-glider* or *to-science folders*

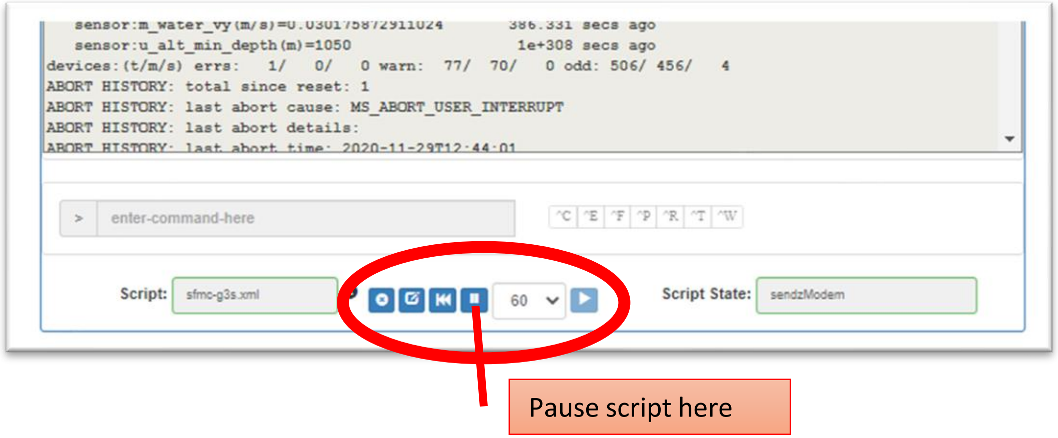
Copies of previously sent files can be downloaded from the *archive* folder

- ❑ Peruse dialog
  - Check values in surface dialog
  - Check device status for warnings and oddities
  - Are files being sent & received?
  - Are ma-files being re-read after sending?
  - Did the call drop out before mission was resumed?
  - Check free space available

- ❑ Manually send commands
  - Only required for sending commands not satisfied by xml script
  - Pause XML script
  - Ctrl-e to extend surfacing time
  - Prepend commands with *!*
  - Ctrl-r to resume mission
  - Play script, rewind to ensure script is in first state

Command Procedures

Surface commands in GliderTerminal	Description
pause xml script	Prevent conflicting commands being sent
Ctrl-w	Print device status
Ctrl-e	Extend surfacing time by 5min
Ctrl-f	Re-read mafiles
Ctrl-t	Or *!consci*, boot into science computer
Ctrl-r	Resume mission
!dockzr *	Send all files from to-glider to the glider, archives sent files.
!type config\sbdlist.dat	Print sbdlist.dat to screen
!sbd load sbdlist.dat	Reload new sbdlist, do not need to do this for new tbdlists
!dockszr *	Send all files (tbdlist.dat in this case) from to-science to the glider scibay, archives sent files.
!type config\tbdlist.dat	Print tbdlist.dat to screen, note that the science computer doesn't *know* its in a mission so no need for ! prepending commands.
quit	Quit science bay
!put u_alt_min_depth 1050	Set sensor to u_alt_min_depth to 1050
!where	Print surface dialog and confirm value of u_alt_min_depth
s *.sbd *.tbd	Send all sbd and tbd files in logs folder
!type mafiles\goto_l10.ma	An example only. Prints goto_l10 to screen and verify file is correct and not corrupted



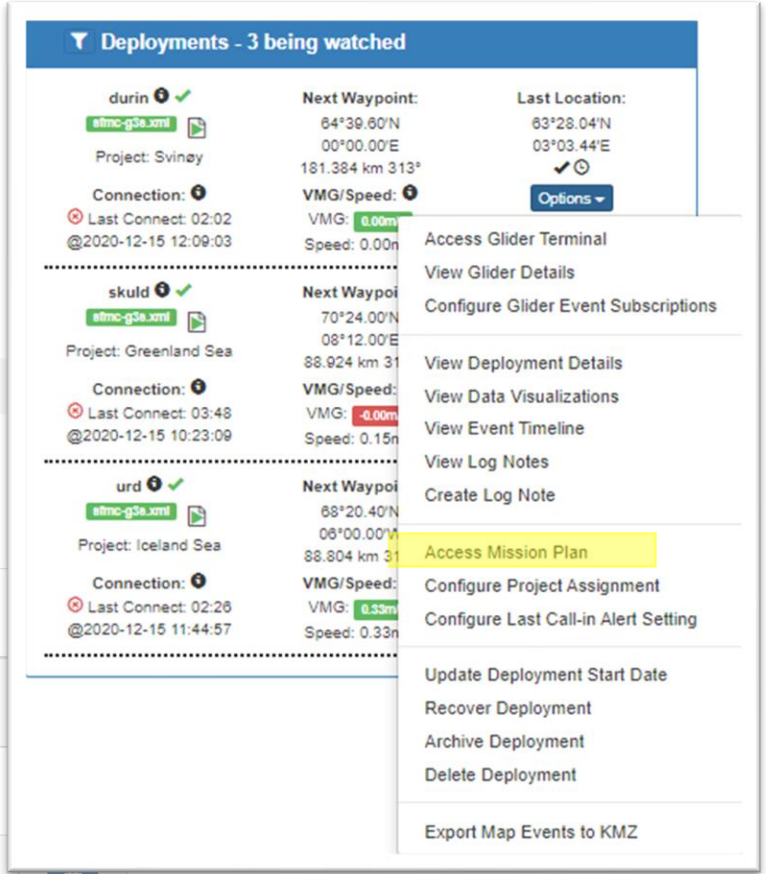
XML Scripts

A list of commands specified in our most used xml scripts

sfmc-g3.xml	sfmc-g3-2.xml	resume.xml	callbackPrimary.xml
s *.sbd *.tbd	s *.sbd *.tbd –num=2	Ctrl-r	h 0 0
!dockszr *	!dockszr *		
!dockzr *	!dockzr *		
Ctrl-f	Ctrl-f		
Ctrl-w	Ctrl-w		
Ctrl-r	Ctrl-r		

Mission Planning Menu

Go to *Configuration>Mission Planning Menu>Mission Plans Menu>Mission Plans*
Or select *Access Mission Plan* from the gliders *Options* dropdown



Mission Plans

Please note that this page does not update in real-time. Perform a refresh to see updates.

Show 15 **Mission Plans**

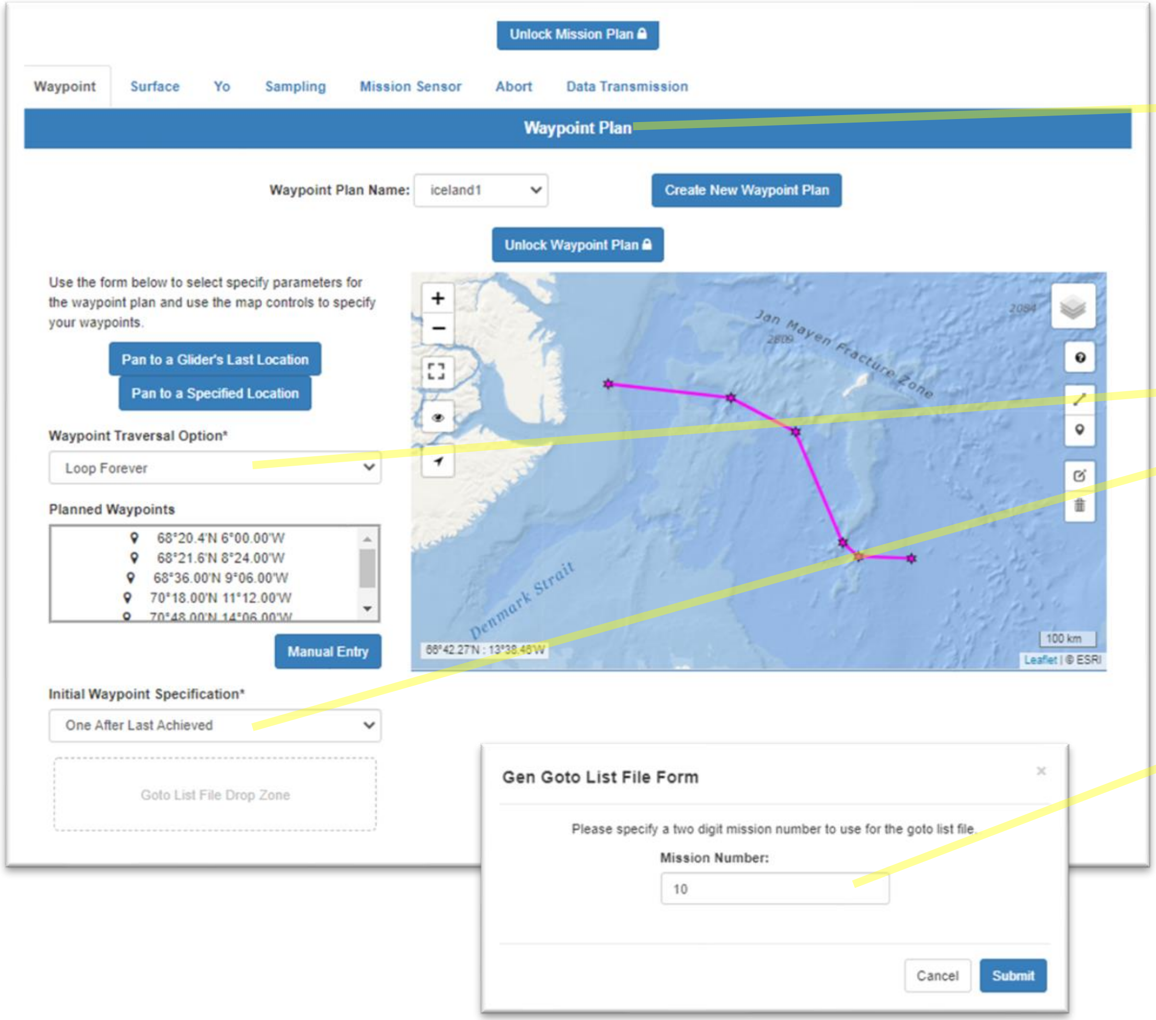
Mission Plan Name	Associated Group	Created By	Creation Date	Last Modified By	Last Modified Date	In Use	View	Edit	Clone
Baffins	bergen	felliott	2021-05-05 09:21	felliott	2021-05-20 09:22:56	No			
Barents	bergen	felliott	2021-05-05 09:19	N/A	N/A	No			
Fram_Strait	bergen	felliott	2021-05-05 09:19	felliott	2021-06-28 07:37:18	Yes			
Gimsøy	bergen	felliott	2021-03-13 07:07	felliott	2021-08-01 09:19:51	No			

CHECKLIST:

- ❑ A lock in the edit field means the mission is locked for editing by the last user that modified, you need to get in touch with this person and get them to unlock it
- ❑ Click *View* to go to the *Mission Plan*

Waypoint Creation

It is best to access waypoint plans via: *Configuration/Mission Planning Menu/Mission Plans Menu/Mission Plans*, then clicking *Edit* for a specific mission. Do not go via the *Options* button > *Access Mission Plan*, there is a bug that will not allow the generation of goto files



CHECKLIST:

- ❑ The *Waypoint* tab is the only tab you will use here
- ❑ Click Edit Mission Plan **Edit Mission Plan**
- ❑ Make sure the correct waypoint plan is selected, usually a *standard transect and click* **Modify Selected Waypoint Plan**
- ❑ A layout similar to this one should be visible
- ❑ Insert or modify waypoints
  - Use buttons on map
  - Drag existing waypoints on map to modify them visually

    or
  - Manually insert waypoints in NMEA format

    or
  - Drag and drop existing goto file in box
- ❑ Select the *Waypoint Traversal Option: Loop Forever*
- ❑ Select the *Initial Waypoint Specification: The one after last achieved*

If no waypoint has been achieved it will default to the closest

- ❑ Click *Unlock Waypoint Plan* The page will reload **Unlock Waypoint Plan**
- ❑ Click *Unlock Mission Plan* (scroll up). The page will reload **Unlock Mission Plan**
- ❑ Scrolling down you will see **Gen Goto List File** click button to generate a new goto file (waypoint list)
- ❑ Input Mission Number of 10, goto_l10.ma will download
- ❑ Navigate to the appropriate glider terminal and drop *goto_l10.ma in the to-glider box*